

Evaluation of real-time performance of local-level flu forecasts for the 2025-2026 season in a subset of locations in the United States

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Our Goal

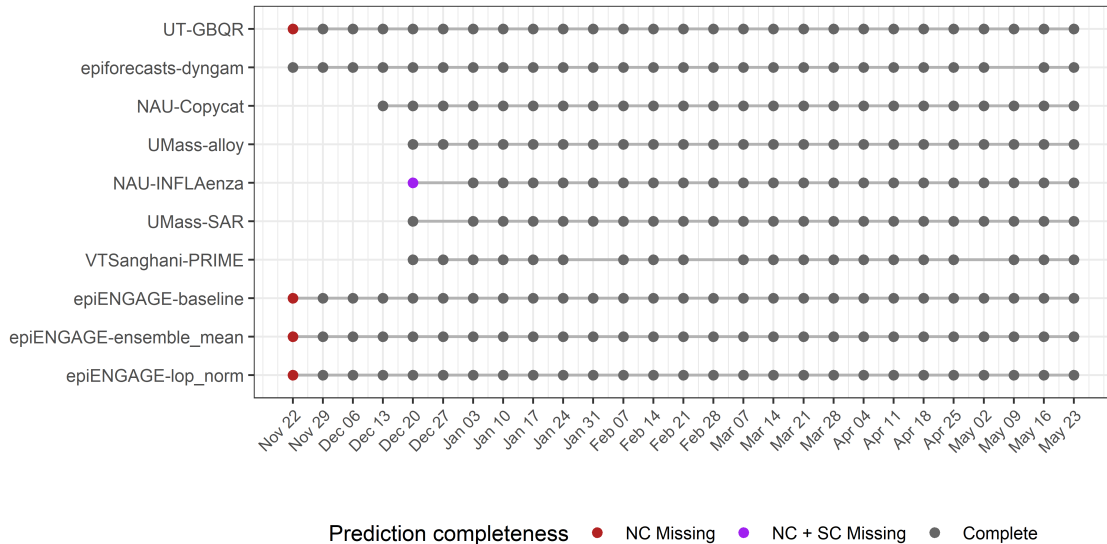
We want to **systematically evaluate forecasts** generated at the local level to answer the following questions:

- ① How does **local level forecast performance compare to the performance of aggregated forecasts** within and across different models, and how do **characteristics of the locality and the aggregate population** impact the relative value of local forecasting?
- ② Which **models/methods perform best** for both local and aggregate locations, accounting for **variables that impact a model's overall forecast performance**?

Analysis Roadmap

	Model Performance	State vs. Local
Forecast Metrics	Weighted Interval Score, 95% Coverage, Scaled Relative Skill Scores <i>[Slides 7-8]</i>	Relative WIS <i>[Slides 9-12]</i>
Regression	Random-Effects GLMM <i>[Slides 17-19]</i>	Random-Effects GLMM with Offset <i>[Slides 13-16]</i>

Timeline of Included Model Submissions



Weighted Interval Score (WIS) and Relative WIS (rWIS)

The **interval score** penalizes **dispersion** ($u_\alpha - l_\alpha$), **underprediction** ($y < l_\alpha$), and **overprediction** ($y > u_\alpha$).

$$\text{IS}_\alpha = (u_\alpha - l_\alpha) + \frac{2}{\alpha}(l_\alpha - y)\mathbf{1}_{(y < l_\alpha)}(y) + \frac{2}{\alpha}(y - u_\alpha)\mathbf{1}_{(y > u_\alpha)}(y) \quad (1)$$

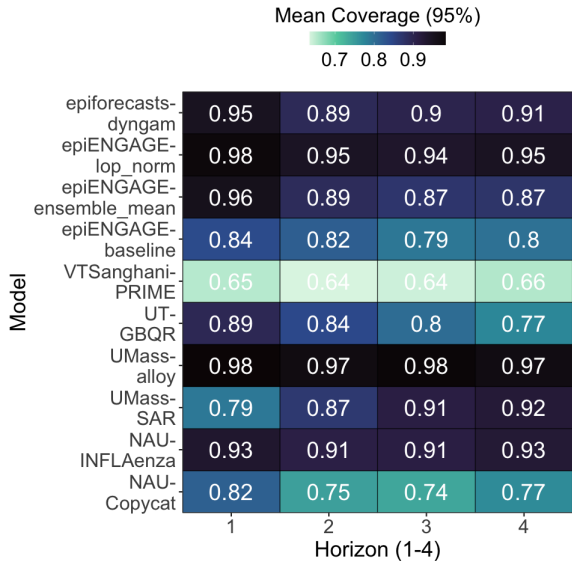
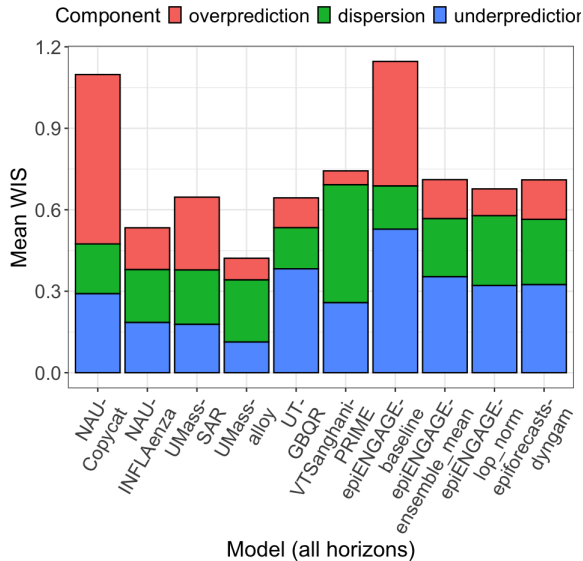
The **weighted interval score** (WIS) is a **weighted average** of the interval scores over K prediction intervals and the absolute error of the median forecast m , with weights $w_k = \alpha_k/2$.

$$\text{WIS} = \frac{\frac{1}{2}|y - m| + \sum_{k=1}^K \frac{\alpha_k}{2} \text{IS}_{\alpha_k}}{K + \frac{1}{2}} \quad (2)$$

The **relative weighted interval score** (rWIS) is computed as the **ratio** of a model's weighted interval score to the baseline weighted interval score. Since lower WIS is better, rWIS of lower than 1 means the **model performs better than the baseline and vice versa**.

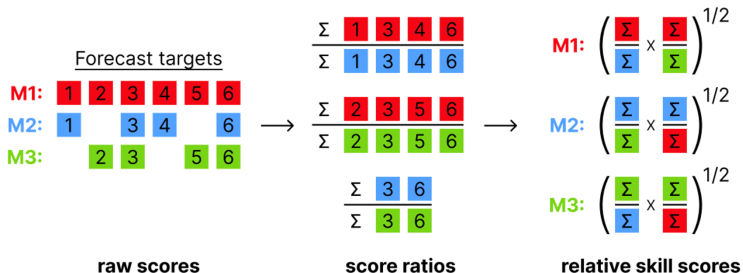
$$\text{rWIS} = \frac{\text{WIS}_{\text{model}}}{\text{WIS}_{\text{baseline}}} \quad (3)$$

WIS component breakdown and 95% coverage per model



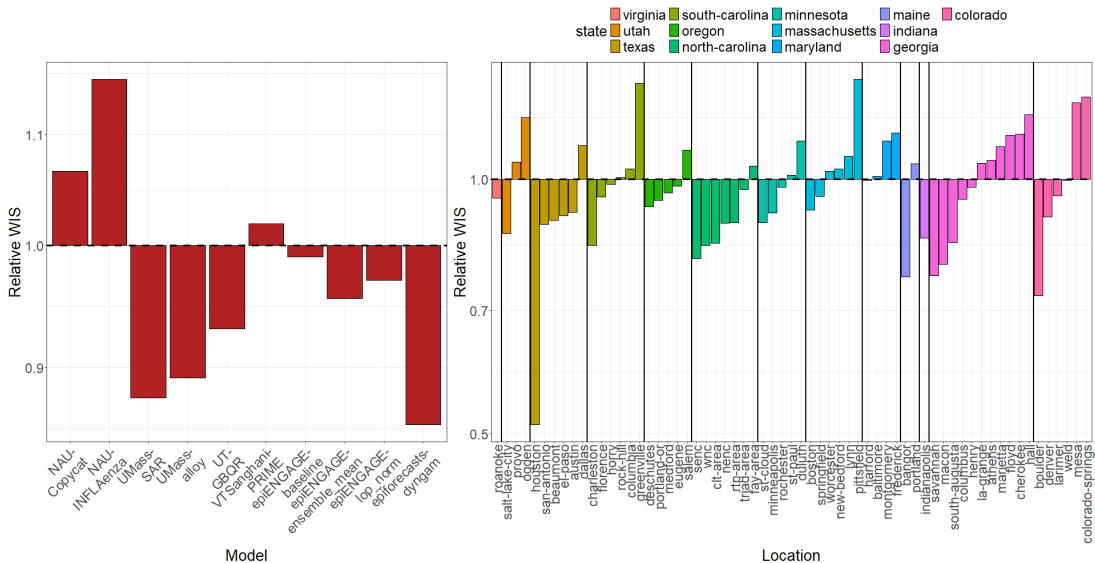
Relative Skill Scores (RSS) and Scaled RSS (SRSS)

A mean score ratio is calculated for every pair of models using only **overlapping forecasts**. The **relative skill score** of a model is the **geometric mean** of all pairwise score ratios involving that model. If a baseline model is available, scores can be **scaled** relative to the baseline. If scaled, then the same principle from rWIS applies where SRSS of lower than 1 means the **model performs better than the baseline and vice versa**.

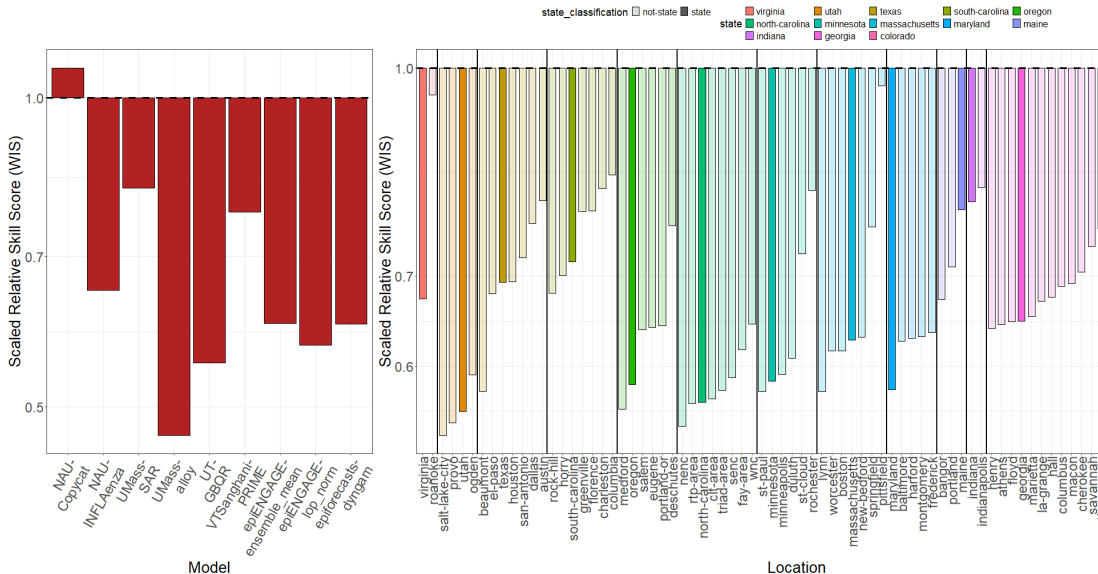


Credit: scoringutils documentation

Relative WIS of models and locations



Scaled Relative Skill Scores (SRSS) of models and locations

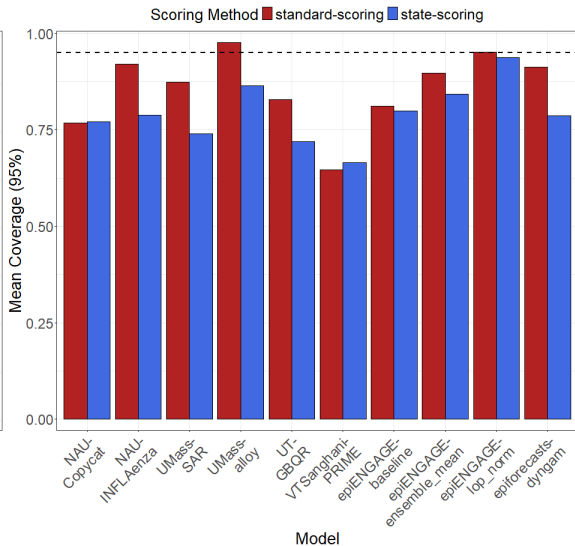
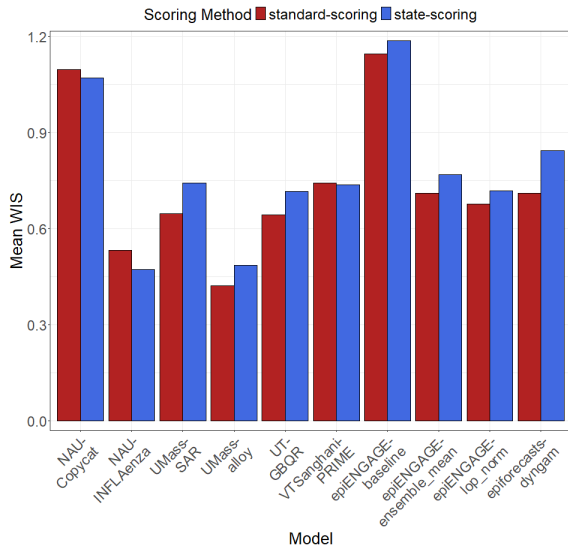


Forecast scoring types

A **standard-scoring forecast** are predictions made by a model on a local (or state) jurisdiction informed by previous observed data of that local (or state) jurisdiction.

A **state-scoring forecast** (on a local jurisdiction) are predictions made by a model on a state jurisdiction informed by jurisdiction observed data of that state jurisdiction.

Standard-scoring vs. state-scoring model performance



Model-based evaluation of aggregate (state) vs. local forecast performance

We first assume **independent and identically distributed errors** on the **log-transformed WIS scores** from the predictions and observations.

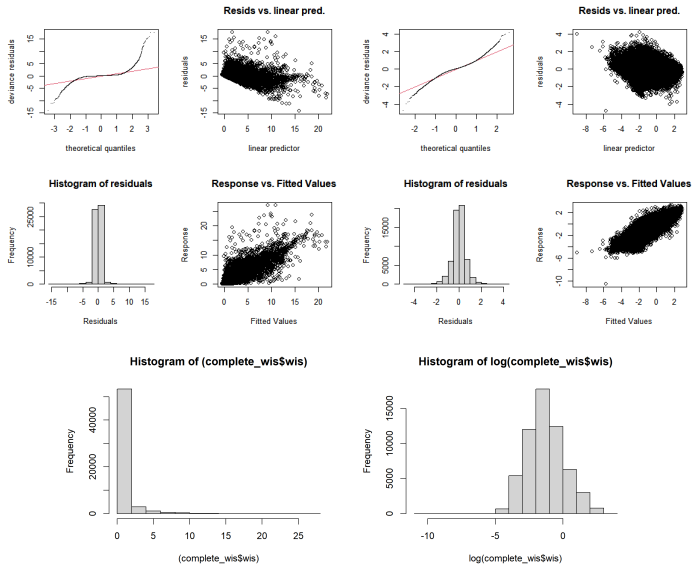
$$\log(\text{WIS}_{h,d,l,m}^{\text{local}}) \sim \text{Normal}(\mu_{h,d,l,m}^{\text{local}}, \sigma) \quad (4)$$

We then model the **expected WIS** of a particular forecast horizon h on forecast date d at location l for model m with a generalized linear mixed model.

$$\mu_{h,d,l,m}^{\text{local}} = \beta_0 + \log(\text{WIS}_{h,d,l,m}^{\text{aggregate}}) + \beta_{1,m} + \beta_{2,l} + \beta_{3,h} \quad (5)$$

where h is the forecast horizon, d is the forecast date, l is the local location of the forecast (the borough or metro area), and m is the model. Location, horizon, and model are modeled as random effects. The $\log(\text{WIS}_{h,d,l,m}^{\text{aggregate}})$ **offset should roughly account for the average forecast difficulty** of a particular forecast location, forecast date, and horizon for the particular model.

Why log-transform?



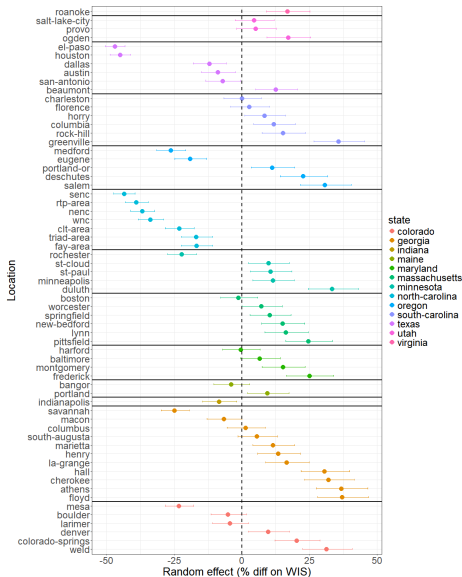
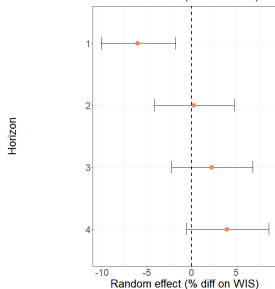
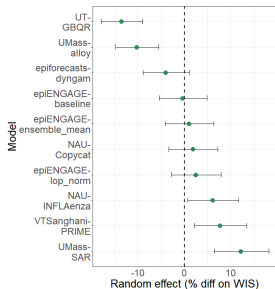
Local vs. aggregate formula in R

This model **log-transforms** both WIS and state WIS (state-scoring WIS) where the **offset** functions as a way to **understand forecast difficulty relative to each specific combination** of date, horizon, location, and model to understand if local does substantially better.

Listing 1. Generalized linear mixed model (GLMM) formula with R package `mgcv`

```
model <- gam(  
  log(wis) ~ # log-transformed WIS  
    offset(log(state_wis)) + # log-transformed state_wis offset  
    s(model_id, bs = "re") + # random effect of 'model_id'  
    s(location, bs = "re") + # random effect of 'location'  
    s(horizon, bs = "re"), # random effect of 'horizon'  
  family = gaussian(), # Gaussian distribution assumption  
  data = complete_wis_filtered, # data source  
  method = "REML" # method with superior stability compared to GCV  
)
```

Some estimated random effects with the offset



Model-based evaluation of model performance

Once again, we will assume we have **independent and identically distributed error** around the **log-transformed WIS scores** on the data.

$$\log(\text{WIS}_{h,d,l,m}) \sim \text{Normal}(\mu_{h,d,l,m}, \sigma) \quad (6)$$

We will model the **expected WIS** of a particular forecast horizon h on forecast date d at location l for model m with a generalized linear mixed model.

$$\mu_{h,d,l,m} = \beta_0 + \beta_{1,m} + \beta_{2,l} + \beta_{3,h} + \beta_{4,t} + \beta_{5,b}, \quad (7)$$

where h is the forecast horizon, d is the forecast date, l is the location of the forecast, m is the model, t is the epidemic trend, and b is the granularity (local or state). Model, location, trend, granularity, and horizon are all modeled as random effects on the log-transformed scores. The epidemic trend of a particular forecast was found by categorizing each target week as "Stable," "Decreasing," or "Increasing" based on the percent change in a trailing 3-week moving average of incidence relative to the prior week's moving average, with changes within 5% classified as "Stable."

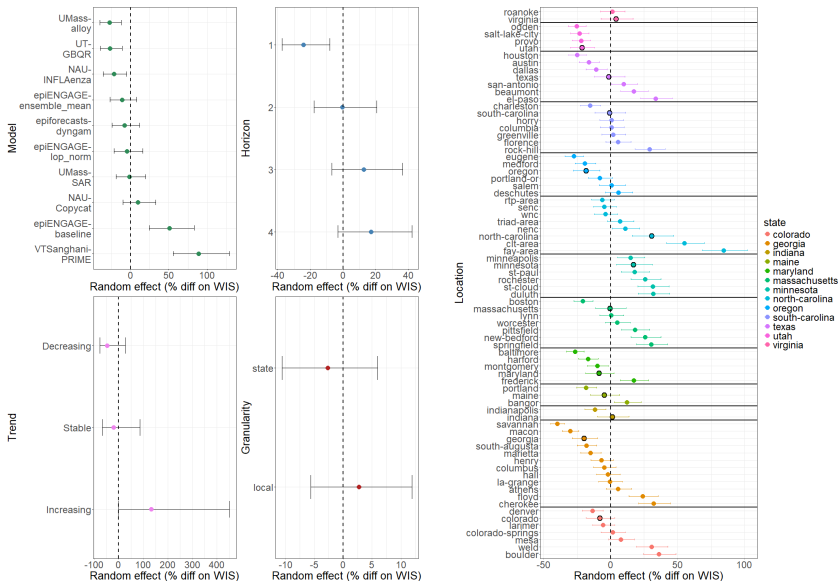
Model performance formula in R

This model log-transforms WIS and then treats the rest of the predictors as **random effects** centered at 0 **that represent deviation from the mean** for each predictor's value.

Listing 2. Generalized linear mixed model (GLMM) formula with R package mgcv

```
model <- gam(  
  log(wis) ~ # log-transformed WIS  
    s(model_id, bs = "re") + # random effect of 'model_id'  
    s(location, bs = "re") + # random effect of 'location'  
    s(horizon, bs = "re") + # random effect of 'horizon'  
    s(trend, bs = "re") + # random effect of 'trend'  
    s(granularity, bs = "re"), # random effect of 'granularity'  
  family = gaussian(), # Gaussian distribution assumption  
  data = standard_wis , # data source  
  method = "REML" # method with superior stability compared to GCV  
)
```

Some more estimated random effects



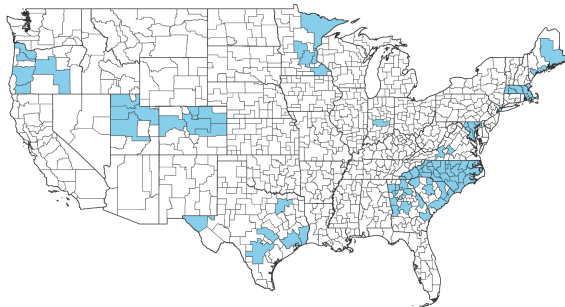
Supplementary Slides

Flu MetroCast Hub

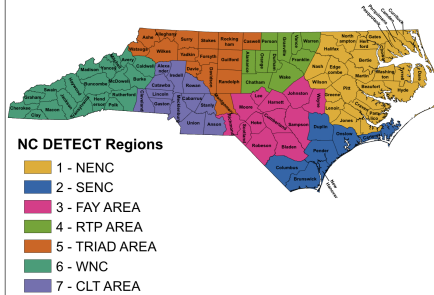
While other forecasting Hubs such as the FluSight Challenge or COVID-19 Forecast Hub are often based at the state and national level, the **Flu MetroCast Hub is designed to focus on the local level**, making it an excellent resource to evaluate.

- The Hub solicited **HSA (Hospital Service Area) or metro-area level predictions and predictions for the corresponding aggregate location** (usually the state).
- The target was mostly **percent of Emergency Department (ED) visits due to flu within an epi-week** and sometimes **percent of ED visits due to Influenza-like-illness (ILI)**. The Hub predictions were in horizons ranging from 1 to 4 weeks (with week 0 being an optional horizon), yet only **scored on horizons 1 to 4**.
- The local jurisdictions were chosen from within the aggregate jurisdiction based on **conversations with public health partners at either the local or jurisdiction-level**, who identified which locations would be of **greatest public health relevance**. Additional considerations were made to ensure **population size was deemed sufficiently large** (> 250,000), though this is not true in all cases.

Hub Jurisdictions



NC DETECT Regions



Jurisdiction

56 HSAs + 12 States
7 NC Regions + State
NYC

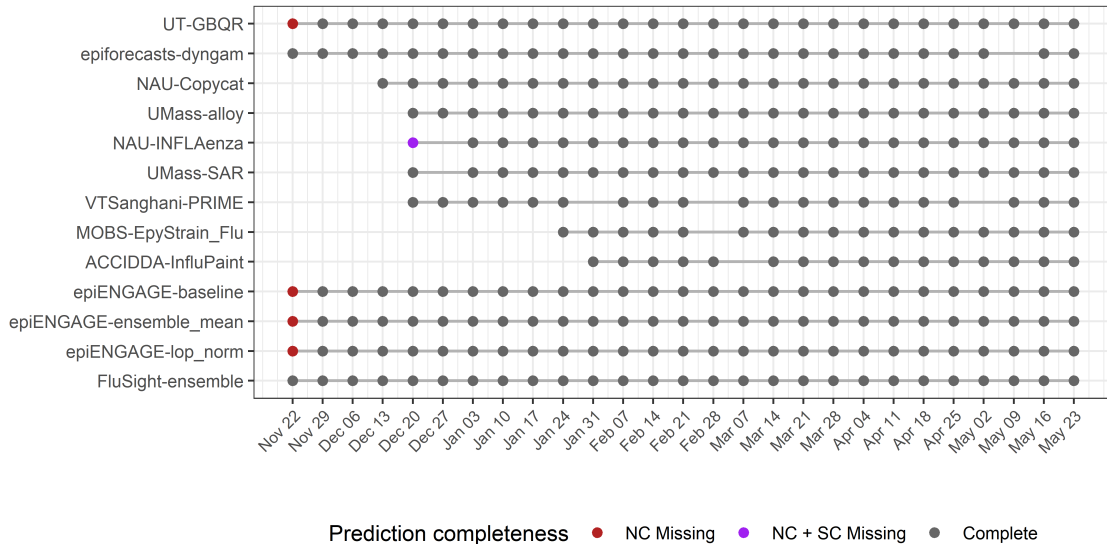
Data Source

CDC NSSP
NC DHHS
NYC DOHMH

Target Outcome

Flu % of ED visits
Flu % of ED visits
ILI % of ED visits

Summary of Model Submissions



Skill score derivation

To define skill scores, we need to start with the parameters of its definition and then build from mean score ratios to relative skill scores to scaled relative skill scores.

We define a group g by the set of combinations it filters by. Those filters can include forecast horizon h , model_id m , scoring approach s , target_end_date d , and location l . The elements of g are set in the code through the `by` parameter. Then, we define the set of possible comparators as C with respect to g , where the elements of C are set through the `compare` parameter.

Skill score derivation cont.

Then, for two comparators $c, c' \in C$, we define $O_{c,c'}^{(g)}$ as the overlap set within group g for which both c, c' have submitted forecasts. Let the target i be defined as before, minus the categories included in C . Then, $\text{WIS}_{c,i}$ is the WIS of comparator c on target i .

Next, we define the mean score ratio. For any c, c' pair, we define the mean score ratio

$$r_{c,c'}^{(g)} = \frac{\sum_{i \in O_{c,c'}^{(g)}} \text{WIS}_{c,i}}{\sum_{i \in O_{c,c'}^{(g)}} \text{WIS}_{c',i}}, \quad (8)$$

where it is a ratio of mean WIS across shared targets.

Skill score derivation cont.

Then, we can let the relative skill score be defined as

$$\text{RSS}_c^{(g)} = \left(\prod_{c' \in C, c \neq c'} r_{c,c'}^{(g)} \right)^{\frac{1}{|C|}}, \quad (9)$$

with $|C|$ as the number of comparators, such that it is the pairwise mean score ratios, with the $\frac{1}{|C|}$ exponent.

Lastly, we write that the scaled relative skill score can be written as

$$\text{sRSS}_c^{(g)} = \frac{\text{RSS}_c^{(g)}}{\text{RSS}_B^{(g)}}, \quad (10)$$

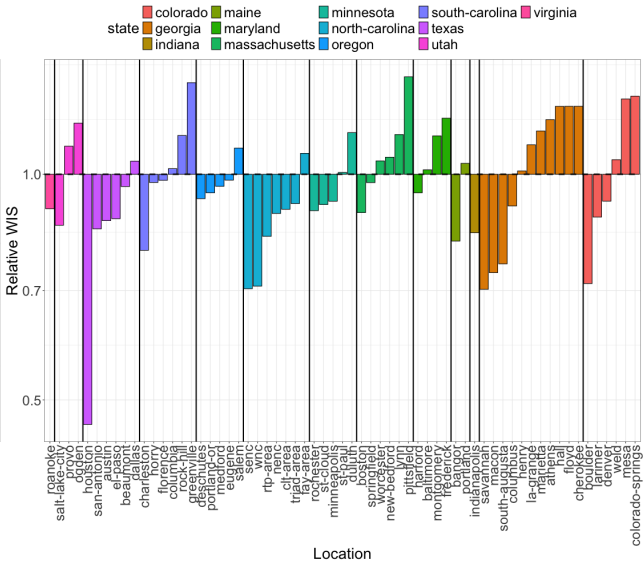
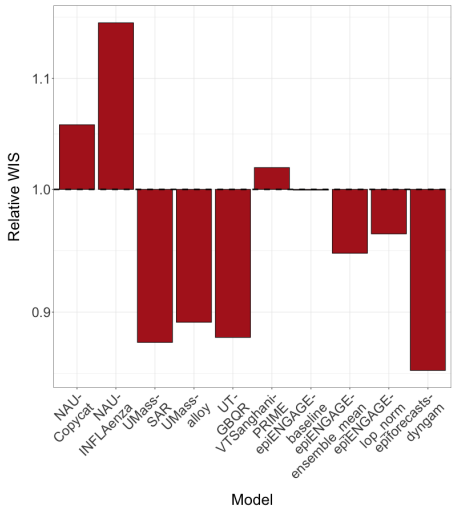
where $\text{RSS}_B^{(g)}$ is the relative skill score of the chosen baseline. The baseline is set using the baseline parameter.

Skill score derivation cont.

Importantly, when $|C| = 2$, which is the case when we are comparing scoring approach, there is only one mean score ratio (ignoring reciprocal duplicates) and thus $\text{sRSS}_c^{(g)} = r_{c,c'}^{(g)}$ since the baseline is set to the state-scoring approach. Therefore, the sRSS reduces to the relative WIS, which is usually calculated as a ratio of mean weighted interval scores (10). Specifically, sRSS becomes relative WIS across overlapping forecasts, where the forecasts only need overlap across the state-scoring and standard-scoring versions of the models.

To use skill scores in our evaluation, we use the `get_pairwise_comparisons()` function from `scoringutils`.

Sensitivity Analysis



Sensitivity Analysis cont.

